

Species-Aware Review: Bacterial Nucleotide Excision Repair (NER) in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu03420` — Nucleotide excision repair (module area: nucleotide_metabolism) **Scope of evidence:** UniProtKB (Swiss-Prot + TrEMBL), KEGG orthology/genome context, and primary literature. All KT2440 protein annotations are homology-inferred (UniProt protein-existence level "Inferred from homology"); no direct KT2440 protein-level enzymatic assays exist in UniProt. Regulatory evidence is direct for KT2440

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1. Executive summary

The **canonical bacterial NER pathway is complete and satisfiable** in *P. putida* KT2440. Every required step of the global-genome (GG-NER) and transcription-coupled (TC-NER) sub-pathways is encoded by a high-confidence gene: damage recognition (**UvrA** PP_0483, **UvrB** PP_1974), dual incision (**UvrC** PP_4098), damaged-oligonucleotide/UvrB turnover (**UvrD** PP_5352), repair-synthesis gap filling (**DNA Pol I** PP_0123), nick sealing (**LigA** PP_4274), and transcription-repair coupling (**Mfd** PP_2148).

Of the 10 genes in the KEGG bucket, **7 are core NER** and **3 are accessory or over-propagated** and should not be counted toward step satisfiability: - **PP_2839** is a **DinG-family Fe-S helicase** placed in `ppu03420` only because KEGG maps it to the *eukaryotic* ERCC2/XPD node (KO K10844). Bacteria have no XPD/TFIIH NER step → **over-propagation / not expected in target taxon**. - **PP_3087** is a **UvrA2 paralog** (shares KO K03701 with true UvrA but lacks NER/SOS GO terms; standalone locus) → **candidate_uncertain / accessory**. - **ligB (PP_4968)** is a **LigB-subfamily** secondary NAD⁺-ligase (KEGG primary pathway = DNA replication) → **candidate_uncertain**; LigA is the functional NER ligase.

No **Cho** alternative 3'-endonuclease is present (single UvrC). The module boundary should exclude the LexA2-controlled **imuABC/dnaE2** translesion-mutagenesis branch, which is a distinct pathway in KT2440.

Overall module verdict: COVERED (all core steps satisfied), with three flagged annotations for cleanup.

2. Target-organism pathway definition

Process included. Bacterial NER is the UvrABC excinuclease system that recognizes and removes **helix-distorting, bulky DNA lesions** (UV cyclobutane pyrimidine dimers and (6–4) photoproducts, intrastrand crosslinks, bulky chemical adducts) by excising a **short 12–13 nt damage-containing oligonucleotide**, followed by repair synthesis and ligation (

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Two entry modes converge on the same UvrBC incision machinery: - **Global-genome NER (GG-NER):** UvrA₂B₂ scans duplex DNA for distortions. - **Transcription-coupled NER (TC-NER):** Mfd recognizes RNA-polymerase stalled at a template-strand lesion, dislodges it, and recruits Uvr(A)BC

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Neighboring pathways to keep separate (do not merge into this bucket): - **Base excision repair (BER)** — glycosylase/AP-endonuclease-initiated removal of small/non-bulky base damage (KEGG ppu03410). Shares no core enzymes with NER. - **Mismatch repair (MMR)** — MutS/MutL/UvrD (ppu03430). Note **UvrD/PcrA (PP_5352)** is shared between NER, MMR, and replication; it is not NER-exclusive. - **DNA replication (ppu03030)** — the primary KEGG pathway for PolA (PP_0123), LigA (PP_4274), and LigB (PP_4968); these are shared enzymes. - **SOS translesion mutagenesis** — LexA2-regulated **imuA-imuB-dnaE2** cassette and TLS polymerases; error-prone lesion *tolerance*, not excision. Must be excluded (

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- **Global "DNA repair" overview maps** should not be conflated with the specific UvrABC excinuclease bucket.

Alternate names / database definitions. UvrABC endonuclease / excinuclease ABC / "excision nuclease" system; KEGG map 03420 is a **combined eukaryote+prokaryote map**, which is the direct cause of the ERCC2/XPD over-mapping (Section 5). GO anchor for the core process: **GO:0006289 nucleotide-excision repair**; TC-NER damage recognition: **GO:0000716**.

3. Expected step model (canonical bacterial NER)

#	Step	Function	Expected enzyme/KO	Present in KT2440?
1	Damage recognition (GG)	UvrA ₂ B ₂ scans DNA; UvrA is ATPase/sensor	UvrA (K03701)	☐ PP_0483
2	Lesion verification / preincision	UvrB melts/probes strand, forms preincision complex	UvrB (K03702)	☐ PP_1974
3	Dual incision	UvrC cuts 3' and 5' of lesion	UvrC (K03703)	☐ PP_4098
4	Oligo + UvrB release	Helicase II unwinds the 12–13 nt fragment	UvrD (K03657)	☐ PP_5352
5	Repair synthesis	Gap filling	DNA Pol I (K02335)	☐ PP_0123
6	Nick sealing	NAD ⁺ -dependent ligation	LigA (K01972)	☐ PP_4274
7	Transcription coupling (TC)	Remove stalled RNAP, recruit UvrAB	Mfd/TRCF (K03723)	☐ PP_2148
—	<i>Alt. 3' incision</i>	Cho endonuclease (some bacteria)	Cho (K19226)	☐ absent (not required)
—	<i>XPD/ERCC2 helicase</i>	eukaryotic TFIIH only	K10844	☐ not applicable to bacteria

All seven core steps are covered. Steps 5 and 6 are enzymatically shared with replication but are the accepted NER repair-synthesis/ligation activities.

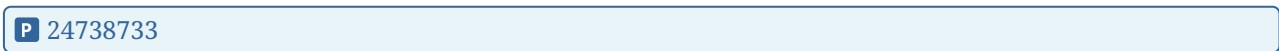
4. Candidate genes and evidence

High-confidence core NER genes (Swiss-Prot, HAMAP-ruled unless noted):

- **uvrA** — PP_0483 / Q88QK7 (944 aa; K03701). UvrABC subunit A, ATPase/DNA-binding damage sensor. Keywords include *SOS response*, *Zinc-finger*; GO:0006289 + GO:0009432. **Role: primary GG-NER damage recognition. Caveat: none — unambiguous. Evidence: homology (strong family assignment).**
- **uvrB** — PP_1974 / Q88LF9 (671 aa). UvrB family; forms preincision complex; GO:0006289. **Role: lesion verification. Caveat: none.**
- **uvrC** — PP_4098 / Q88FJ7 (607 aa). UvrC family; N-terminal 3' incision, C-terminal 5' incision; GO:0006289. **Role: dual incision. Caveat: single copy — no Cho backup.** (*Note: neighboring PP_4099 "uvrY" is actually GacA response regulator — a gene-name artifact, NOT NER.*)
- **mfd** — PP_2148 / Q88KZ1 (1149 aa; K03723; EC 5.6.2.4). TRCF; N-terminal UvrB-homology module + C-terminal RecG-like helicase; GO:0000716. **Role: transcription-coupled NER. Caveat: none; well-conserved.**
- **uvrD** — PP_5352 / Q88C31 (728 aa; K03657; EC 5.6.2.4). UvrD/helicase II subfamily (TrEMBL). **Role: unwinds excised oligo + displaces UvrB. Caveat: pleiotropic — also functions in MMR and replication; not NER-exclusive.**
- **polA** — PP_0123 / Q88RK6 (915 aa; K02335; EC 2.7.7.7; TrEMBL). Pol I with 5'→3' and 3'→5' exonuclease. **Role: NER repair synthesis. Caveat: primary role/KEGG pathway is replication; shared enzyme.**
- **ligA** — PP_4274 / Q88F25 (776 aa; K01972; EC 6.5.1.2; LigA subfamily). NAD⁺-dependent ligase, "essential for DNA replication and repair." **Role: NER nick sealing. Caveat: essential, shared with replication; this is the functional NER ligase.**

Accessory / flagged genes:

- PP_2839 / Q88J10 (762 aa). KEGG KO K10844 (ERCC2/XPD, EC 5.6.2.3); UniProt = **helicase, DinG subfamily**, 4Fe–4S cluster; GO only 0006281 (generic DNA repair). Standalone locus. **Assessment: DinG R-loop helicase**

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over-propagated to NER via the eukaryotic XPD node.

- PP_3087 / Q88IB2 (838 aa). KO K03701 (same as UvrA) but **UvrA2 paralog**: TrEMBL, no SOS keyword, no GO:0006289 (only 0006281). Standalone, adjacent to ECF sigma-70 factor

PP_3086 (not a uvrA2B2 operon). **Assessment: accessory/stress-associated UvrA paralog; not the primary sensor.**

- **ligB — PP_4968 / Q88D59** (566 aa; K01972; LigB subfamily MF_01587). KEGG primary pathway = DNA replication (ppu03030). **Assessment: secondary ligase of uncertain function; LigA is the NER ligase.**

Evidence type across all genes: homology-based (UniProt "Inferred from homology"); family/KO assignments are unambiguous for the 7 core genes. Regulatory embedding (SOS/LexA1) is directly demonstrated in KT2440 (P 17933893).

5. Gaps, ambiguities, and likely over-annotations

- **Over-propagation #1 (structural artifact of KEGG's combined map): PP_2839 → ERCC2/XPD.** KEGG map 03420 contains eukaryotic TFIIH nodes; the bacterial DinG helicase (KO K10844) is auto-slotted there. Bacterial UvrABC NER has **no XPD/ERCC2/TFIIH step**. DinG's real characterized function is **R-loop resolution and Fe–S redox coordination with BER** (P 24738733).
→ Mark the XPD/ERCC2 step **not_expected_in_target_taxon**; PP_2839 is **not** an NER step.
- **Paralog over-count #2: PP_3087 (UvrA2).** Sharing KO K03701 inflates the UvrA step to two genes. Only PP_0483 carries the NER/SOS annotation signature; PP_3087 lacks it and sits in a non-operonic, sigma-factor-adjacent context. → Keep **one** covered UvrA (PP_0483); PP_3087 = **candidate_uncertain**.
- **Enzyme-family over-count #3: ligB (PP_4968).** Two LigA/LigB genes share KO K01972. LigA (essential) is the NER ligase; LigB is a secondary ligase whose KEGG primary bucket is replication. → Keep **one** covered ligase (LigA); ligB = **candidate_uncertain**.
- **Shared-enzyme caveats:** UvrD, PolA, and LigA are pleiotropic (MMR/replication). They legitimately serve NER but should be cross-annotated, not treated as NER-exclusive markers.
- **True gap check — none critical.** Cho (alternative 3' endonuclease) is **absent**, but Cho is dispensable where UvrC is intact, so this is **not a gap**. No core step is missing.
- **Boundary error to avoid:** do not absorb the LexA2 **imuA–imuB–dnaE2** SOS-mutagenesis cassette or TLS polymerases into this bucket — that is a separate lesion-tolerance pathway

in KT2440

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6. Module and GO-curation recommendations

Per-step module status:

Module step	Gene	Status
UvrA damage recognition	PP_0483	covered
UvrB verification	PP_1974	covered
UvrC dual incision	PP_4098	covered
UvrD oligo release	PP_5352	covered (shared enzyme)
Pol I repair synthesis	PP_0123	covered (shared enzyme)
LigA nick sealing	PP_4274	covered (shared enzyme)
Mfd TC-NER coupling	PP_2148	covered
UvrA2 paralog	PP_3087	candidate_uncertain (accessory)
Secondary ligase	PP_4968 (ligB)	candidate_uncertain
ERCC2/XPD (DinG)	PP_2839	not_expected_in_target_taxon (over-propagated)

Module verdict: COVERED — the module is satisfiable without any of the three flagged genes.

Module boundary / revision: The generic KEGG bucket is broadly correct but imports a **eukaryote-only step (XPD/ERCC2)** that is meaningless for bacteria — **module_needs_revision** to (a) drop the ERCC2/XPD node for prokaryotic taxa, and (b) demote PP_3087/ligB from primary NER membership.

GO-curation actions: - **Remove/avoid GO:0006289** propagation to **PP_2839** (DinG); prefer R-loop/helicase terms (e.g., GO:0004386 helicase activity, GO:0051880/GO:0000737-type processes as appropriate) — flag for manual review. - **PP_3087:** retain generic GO:0006281 unless experimental UvrA2 evidence appears; do not assign GO:0006289 by paralogy alone. - **PP_4968 (ligB):** keep GO:0006281/0006260; do not assert NER-specific GO without evidence. - Core seven

genes: GO:0006289 (and GO:0000716 for Mfd) assignments are appropriate. - No new module document is required, but a **prokaryote-specific NER module** (UvrABCD + PolA + LigA + Mfd, excluding XPD/ERCC2) would prevent recurrent over-propagation across Gram-negative genomes.

7. Genes to promote to full fetch-gene review

1. **PP_2839 (DinG helicase)** — **highest priority**. Resolve the ERCC2/XPD mis-mapping; confirm DinG identity and reassign out of NER. Directly affects module boundary correctness.
 2. **PP_3087 (UvrA2)** — verify paralog status, operon context, and any stress/resistance phenotype; decide accessory vs. redundant.
 3. **PP_4968 (ligB)** — confirm LigB-subfamily assignment and non-NER primary role; keep as secondary ligase.
 4. *(Optional)* **PP_5352 (uvrD)** — TrEMBL entry shared across NER/MMR; a review would document multi-pathway membership cleanly.
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8. Key references

- P 35149830
— Seck et al. (2022), *Nucleic Acids Res.*-adjacent. In vitro reconstitution of *Deinococcus radiodurans* UvrABC NER; defines the sequential UvrA/B/C action releasing a 12–13 nt fragment. *(Related-organism mechanism; strong transfer to KT2440 for core steps.)*
- P 33480355
— Kang et al. (2021). Cryo-EM of Mfd disrupting stalled transcription complexes; Mfd removes stalled RNAP and recruits Uvr(A)BC. *(Mechanistic; strong transfer.)*
- P 38513450
— Kaja et al. (2024). Single-molecule test of Mfd- vs UvrD-dependent TC-NER models in *E. coli*. *(Mechanistic context for coupling factors.)*

- P 24738733**

— Grodick et al. (2014), *Cell*. DinG is an ATP-dependent, 4Fe–4S R-loop-repair helicase coordinating with EndoIII via DNA charge transport. (*Basis for excluding PP_2839 from NER.*)
- P 17933893**

— Abella et al. (2007), *J. Bacteriol.* **Direct KT2440 evidence:** dual LexA regulons; LexA1 = conventional SOS response, LexA2 = imuA/imuB/dnaE2 mutagenesis cassette. (*Defines module boundary and regulatory context.*)

• **Databases:** UniProtKB entries Q88QK7, Q88LF9, Q88FJ7, Q88KZ1, Q88C31, Q88RK6, Q88F25 (core); Q88J10, Q88IB2, Q88D59 (flagged). KEGG **ppu03420** and KO assignments K03701/K03702/K03703/K03657/K02335/K01972/K03723/K10844.

Limitations

- All KT2440 protein annotations are homology-inferred; no strain-specific enzymatic or knockout data for NER components were located in this review (the strongest KT2440-direct evidence is regulatory, **P 17933893**).
- UvrB/UvrC KO numbers (K03702/K03703) are stated from canonical NER conventions; the KEGG bucket was verified for all 10 candidates but per-gene KO for *uvrB/uvrC* was not separately re-queried.
- UvrA2 (PP_3087) function is inferred from the general bacterial UvrA2 pattern and local genomic context, not from KT2440 experiments.

Future experiments / expert questions

- KT2440 knockout/UV-survival phenotyping of *uvrA* (PP_0483) vs *uvrA2* (PP_3087), and of *ligA* vs *ligB*, to resolve functional redundancy.
- ChIP/transcriptomics to confirm LexA1-dependent SOS induction of *uvrA/uvrB/uvrD* in KT2440.
- Biochemical confirmation that PP_2839 is a DinG R-loop helicase (Fe–S, 5'→3') and lacks UvrABC-associated activity.

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